



ToneFill generates seamless dialogue room tone. It listens to a production clip, finds the cleanest, steadiest background ambience, and rebuilds it into a continuous bed you can stretch under any edit — no loop points, no audible joins. VST3 and Audio Unit (both with ARA) for Nuendo, Cubase, Reaper, Studio One and Logic Pro, and a native AAX AudioSuite build for Pro Tools.

■ QUICK START

- 1 Add ToneFill to a dialogue clip. In ARA hosts (Nuendo, Cubase, Reaper, Logic) it is an **Extension** on the clip, not a track insert; in Pro Tools it is an **AudioSuite** plug-in. Analysis runs off the audio thread.
- 2 Leave **Auto** on — the waveform lights the clean ambience it found in **orange**.
- 3 Play back (ARA) or press **Preview** (AudioSuite): the gaps and the whole region are filled with matched room tone.
- 4 Not clean enough? Raise **Voice Reject** and lower **Clean Level**.
- 5 Set **Export Len** and hit **Export WAV** (ARA), or **Render** the clip (AudioSuite), for a standalone bed.

■ DETECTION — WHAT COUNTS AS ROOM TONE

Clean Level

Auto

How quiet a moment must be to count as room tone. Lower = stricter — rejects claps, clicks and loud bits more aggressively. Very high values can pick digital silence between words, giving a silent bed — keep it moderate.

Voice Reject

Auto

How hard to exclude voice, breaths and mouth noise. Higher = cleaner tone. Push it up and keep Clean Level low for dialogue-heavy takes.

Min Fill

Auto

Minimum length a stable fragment must have to be used. Longer = fewer, more consistent chunks and fewer audible joins. Range 0.2 – 5.0 s

■ STRUCTURE — HOW THE BED IS ASSEMBLED

Flatness

Selection

How strict the stationarity requirement is. Higher = only very steady stretches, so the fill avoids audible jumps and level swings.

Bands

Spectral

Spectral mode only: how many frequency bands the source is split into. Fewer = wider bands; more = narrower, finer bands. Grey unless Spectral is active.

Crossfade

Bed

How much neighbouring chunks overlap and blend. Higher = smoother joins. Joins are also widened automatically when two chunks differ in the low end.

■ TEXTURE — OUTPUT SHAPING

Smoothness

Enhance

Enhance only: resynthesis window size. Higher = smoother, more diffuse; lower keeps more of the source's fine texture.

Output

Level

Output volume in dB. Bypassed while Normalize is on. Range -24 – +24 dB

Export Len

Export

Length of the WAV written by Export. The seamless loop is tiled to exactly this length. Range 0.5 – 30 s

■ MODE SWITCHES

Auto / Manual

Auto finds clean room tone for you. **Manual** lets you drag your own regions on the waveform below — drag to add, click a region to remove.

Enhance / Full

Enhance turns on PaulStretch resynthesis for when the real fragments won't blend into a clean bed (enables Smoothness and the Hiss Filter). **Full** analyses the whole take instead of the first 10 minutes — use it when clean tone is scattered across a long clip.

Classic / Experimental /

Spectral

Classic is the tuned selection engine (default). **Experimental** scores every frame with a weighted model. **Spectral** is the per-band mosaic — see below. Pick one; A/B by ear.

Bypass

A/B monitor: play the original source instead of the room tone, so you can compare them. Does not change what Export/Render writes.

■ SPECTRAL MODE & THE BAND EDITOR

Spectral stops looking for one moment that is clean across the whole spectrum. Instead it splits the source into frequency bands, finds clean room tone in *each band on its own*, and recombines them. Because room tone is far steadier within a narrow band, the result has a rock-solid low end and more usable material — the low band gets its own dedicated selection rather than needing every frequency clean at the same instant. The **Bands** knob sets how many bands.

Advanced opens the **Spectral Band Editor**: a spectrogram of the source (log-frequency, colour-mapped) with a draggable line for every band edge. Drag a line to place that band's frequency boundary by hand on the actual spectrum. Zoom with the H / V buttons, pan with the wheel (up/down) and side-wheel (left/right) or by dragging the canvas, and use **Speed** and **Bright** to tune the navigation and readability. Your edges are saved with the clip.

Auto (in the band editor) lets ToneFill choose the layout for you. It runs a proper analysis of the *clean* room tone this clip already selected — a long-term average spectrum plus how steady each frequency is over time — finds where the room's tone concentrates (mains hum, HVAC, resonances, broadband floor) and how many such regions there are, and places the band edges in the quiet valleys between them. The

Bands knob then follows the count it found. It is a starting point, not a lock: drag any edge afterwards to fine-tune.

■ HISS FILTER (Enhance only)

Enhance can add a faint high-frequency hiss as it resynthesises the bed. The Hiss Filter is a live low-pass — applied in real time, so you hear the change instantly with no re-render.

Freq

3–15 kHz

The corner frequency. Lower = more aggressive HF removal. Start around 9 kHz and pull down until the hiss goes.

Q

0.3–2.0

The slope / resonance at the corner. Around 0.7 is a clean, gentle roll-off; higher tightens the knee.

Tip. The switch is only active while Enhance is on — with Enhance off, the real-audio bed already keeps the source's high end.

■ LOUDNESS — NORMALIZE

Turn **Normalize** on to bake the output to a fixed loudness target. Pick **LUFS** (integrated, EBU R128) or **dBFS** (peak), set the target with the – / + stepper, and the **now** readout shows the measured value. While Normalize is on, the Output knob is bypassed and the exported bed is corrected to hit the target precisely at your chosen length (true peak capped at –1 dBFS).

■ WAVEFORM & READOUTS

The bottom panel shows the whole analysed clip. **Orange** bars are the **ambience ToneFill** selected; grey is rejected. In Manual mode, drag across the waveform to add room-tone regions and click one to remove it. **Expand** opens a large window with a timecode ruler, zoom and scroll for precise selecting.

The status line reads:

used 25.1 s (7 chunks) · avail 83.9 s · seam 1.9 dB · in -66 dB

used	total clean material actually used after the Min Fill filter, and how many chunks.
avail	total clean material found before Min Fill — if this is high but <i>used</i> is low, lower Min Fill.
seam	average spectral mismatch across chunk joins (dB). A Δ over 4 dB means raise Min Fill or tighten the selection.
in	the analysed source level. If it reads ≈ -120 dB the selection is picking silence — lower Clean Level.

Pro Tools. AudioSuite recomputes only during **Preview / Render**, so the waveform and readouts refresh while a *recomputing...* hint is showing — keep Preview running while you tweak in Auto.

■ ACTIONS

Regenerate rolls a new random variation of the fill with the same settings — handy if a particular realisation repeats audibly. **Export WAV** writes the fill to a 24-bit WAV, tiled seamlessly to **Export Len** (loudness-corrected when Normalize is on). In Pro Tools, **Render** commits the AudioSuite result to the clip instead.

■ WORKING THE LOW END

Room tone lives or dies below 500 Hz — that's where the ear hears a "different room". ToneFill treats the low band as a first-class part of the room's colour: chunks with a mismatched low end are rejected during selection, and joins between chunks that differ in the low end are crossfaded harder. If a fill still feels like it shifts weight at a cut, raise

Min Fill and **Flatness**, or switch to **Spectral** so the low band gets its own dedicated selection.

■ PER-CLIP SETTINGS

In ARA hosts, every clip keeps its *own* ToneFill settings — knobs, mode, Spectral bands and the hand-drawn band edges are stored per clip and saved in the project, so switching between clips or reopening a session restores exactly what each one had. Attaching ToneFill to a new clip starts it from the defaults, not from the previous clip.

■ FORMATS & INSTALL

VST3 (with ARA)	/Library/Audio/Plug-Ins/VST3/ToneFill.vst3 — Nuendo, Cubase, Reaper, Studio One. Add as an ARA Extension on the clip.
Audio Unit (with ARA)	/Library/Audio/Plug-Ins/Components/ToneFill.component — Logic Pro.
AAX (AudioSuite)	/Library/Application Support/Avid/Audio/Plug-Ins/ToneFill.aaxplugin — Pro Tools, under AudioSuite.
Installer	Signed macOS .pkg with a Customize pane to pick AAX / VST3 / AU; a signed .exe for Windows (VST3).
Manufacturer	cactuzz sound